Proof

- LHS is the number of bitstrings of length (n+1) that contain exactly (r+1) 1's.
- RHS: Consider the possible locations of the final 1 in a string with $(r+1)$ 1's. This final 1 must occur at location $r+1, r+2, \ldots$, or $n+1$. Furthermore, when the last 1 is the kth bit $(r+1 \leq k \leq n+1)$, there must be r 1s among the first $k-1$ locations. Consequently, there are $\binom{k-1}{r}$ such bit strings. Summing over k, the number of bit strings of length $n+1$ containing exactly $r+1$ ones is:

$$\binom{n+1}{r+1} = \sum_{k=r+1}^{n+1} \binom{k-1}{r} = \sum_{j=r}^{n} \binom{j}{r}$$

Because both sides of the identity count the same objects, they must be equal. This completes the proof.